

Data Collection Strategies

Three channels: web/existing datasets, synthetic generation, human annotation — with quality trade-offs

9 min

Duration

B3 – Apply

Bloom's

L2 – Intermediate

Level

AI-FT-IN-TH-000015

ID

Channel 1: Web Scraping & Existing Datasets

General Corpora

Common Crawl, The Pile, RedPajama, C4. Billions of documents. Requires deduplication, toxicity filtering, quality scoring.

Domain Corpora

PubMed (biomedical), arXiv (scientific), GitHub (code), SEC filings (financial). High domain signal but requires significant cleaning.

Licensing

Check licenses carefully: Common Crawl (unknown), The Pile (mixed), Dolly (Apache 2.0). For commercial products: use only permissive licenses.

Scale vs Quality

Web data is plentiful but noisy. Even with filtering, ~5–10% of web-scraped examples are low quality. Acceptable for pre-training; risky for fine-tuning.

Channel 2: Synthetic Data — Self-Instruct / Alpaca Pipeline



Channel 3: Human Annotation — Production Workflow

Annotation Workflow

- Write annotation guidelines (20–50 page document)
- Recruit annotators via Scale AI, Surge AI, Prolific
- Run pilot with 50 examples + calibration session
- Measure inter-annotator agreement ($\kappa > 0.7$ target)
- Blind review of 10% of annotations for QA
- Dispute resolution for conflicting labels

Quality Assurance

- Inter-annotator agreement: Cohen's κ or Krippendorff's α
- Gold standard set: 100 pre-labelled examples for annotation accuracy
- Annotator performance tracking: flag low-accuracy annotators
- Iterative guideline refinement based on common disagreements
- Cost: \$0.50–\$5.00 per example (human) vs \$0.01 (GPT-4)

KEY REFERENCES

- Wang, Y. et al. (2022). Self-Instruct. ACL 2023.
- Scale AI. scale.com
- Laurer, M. et al. (2023). Llama-based instruction fine-tuning. arXiv.

 **Next: Concept 26: Data Preprocessing & Cleaning (AI-FT-IN-TH-000016)**